

# ACAD 245

# DESIGNING PRODUCTS I

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A. BLUETOOTH SPEAKER  
B. HIKING SHOE  
C. LIGHT

SAMA SHAH  
SPRING 2024

# PROJECT A: **BLUETOOTH SPEAKER**

*a good quality speaker that is heavy duty and portable.*





# BACKGROUND & RESEARCH

Target Customer: *College Students*

Age: 18-25 Yr.

*young adults who live in close proximity who enjoy listening to music in social situations, on their own time, or during activities such as going to the beach, hiking, etc.*





# COMPETITOR PRODUCTS

Rating: 3.5/5



Air Audio Speaker (\$219)

PROS: highly modular

CONS: only 7.5 hrs battery, only water resistant, poor sound quality and quiet

Rating: 4.4/5



B&O A1 (\$200)

PROS: 18 hr battery life, water & dust proof, great sound quality

CONS: expensive, complaints about app

Rating: 4.7/5



Bose SoundLink Micro (\$120)

PROS: extremely portable, water & dust proof, very small

CONS: only 6 hrs battery, ok sound quality

Rating: 4.8/5



JBL Clip 4 (\$80)

PROS: 10 hr battery life, water & dust proof, very small/portable

CONS: ok sound

Rating: 4.5/5



Boom 3 (\$117)

PROS: 15 hr battery life, water & dust proof, 360° sound, can pair multiple

CONS: micro-usb charging, manufacturing charging issues, ok sound

Rating: 4.8/5



JBL Flip 5 (\$90)

PROS: 12 hr battery, many designs, water proof, able to pair w multiple speakers

CONS: lacking in sound quality and sound stage, unreliable battery life,

Rating: 4.8/5



JBL PartyBox 310 (\$440)

PROS: large sound, flashy lights, 18 hr battery life

CONS: huge size, expensive, not waterproof

N/A



SIMO X (\$180)

PROS: modular (splits into 4), decently loud

CONS: only 8hr battery life, not waterproof

Rating: 4.6/5



Anker Soundcore (\$22)

PROS: 24 hr battery, many designs, super affordable

CONS: lacks bass & volume

Rating: 4.7/5



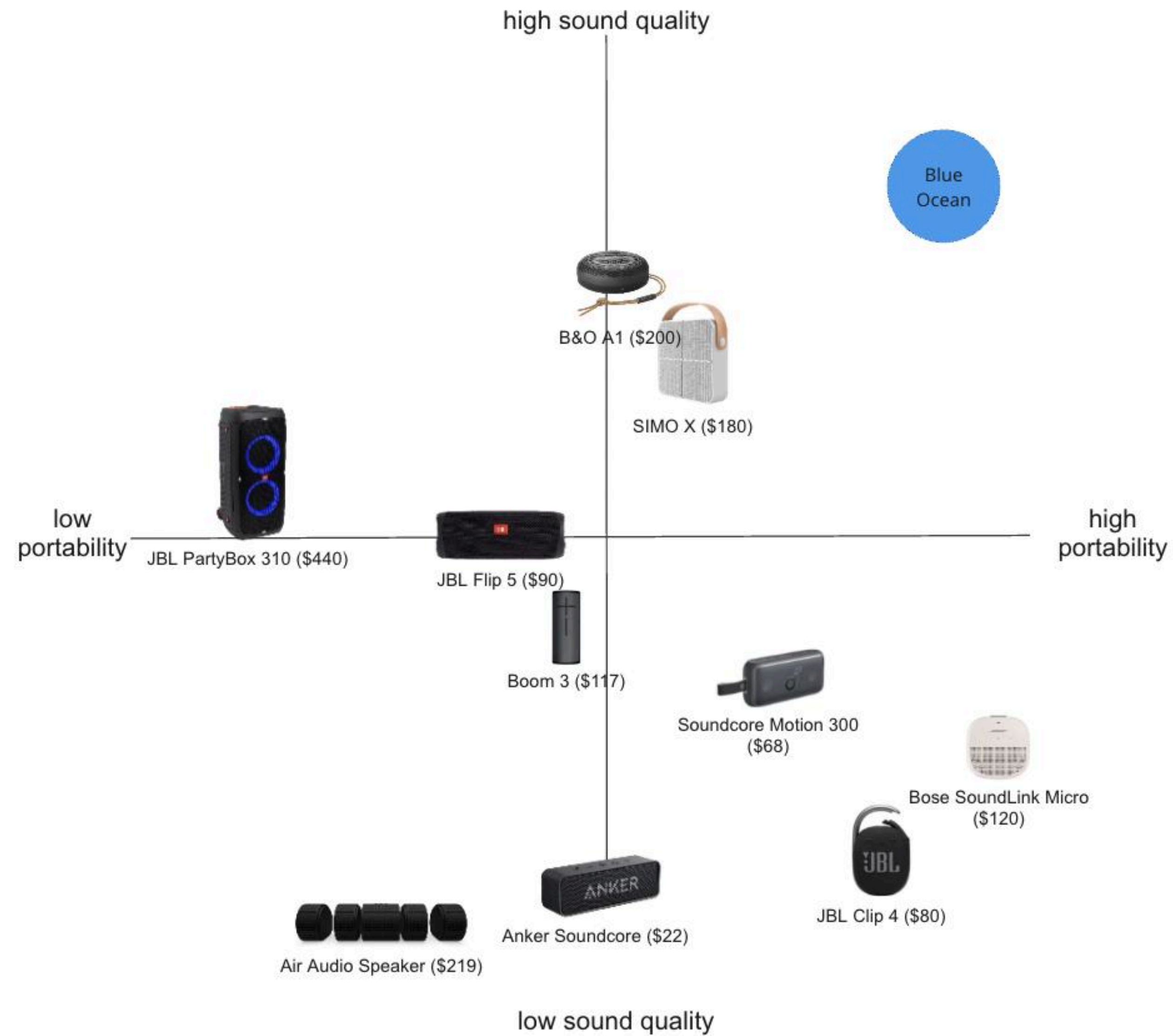
Soundcore Motion 300 (\$68)

PROS: waterproof, 13 hr battery life, portable, affordable

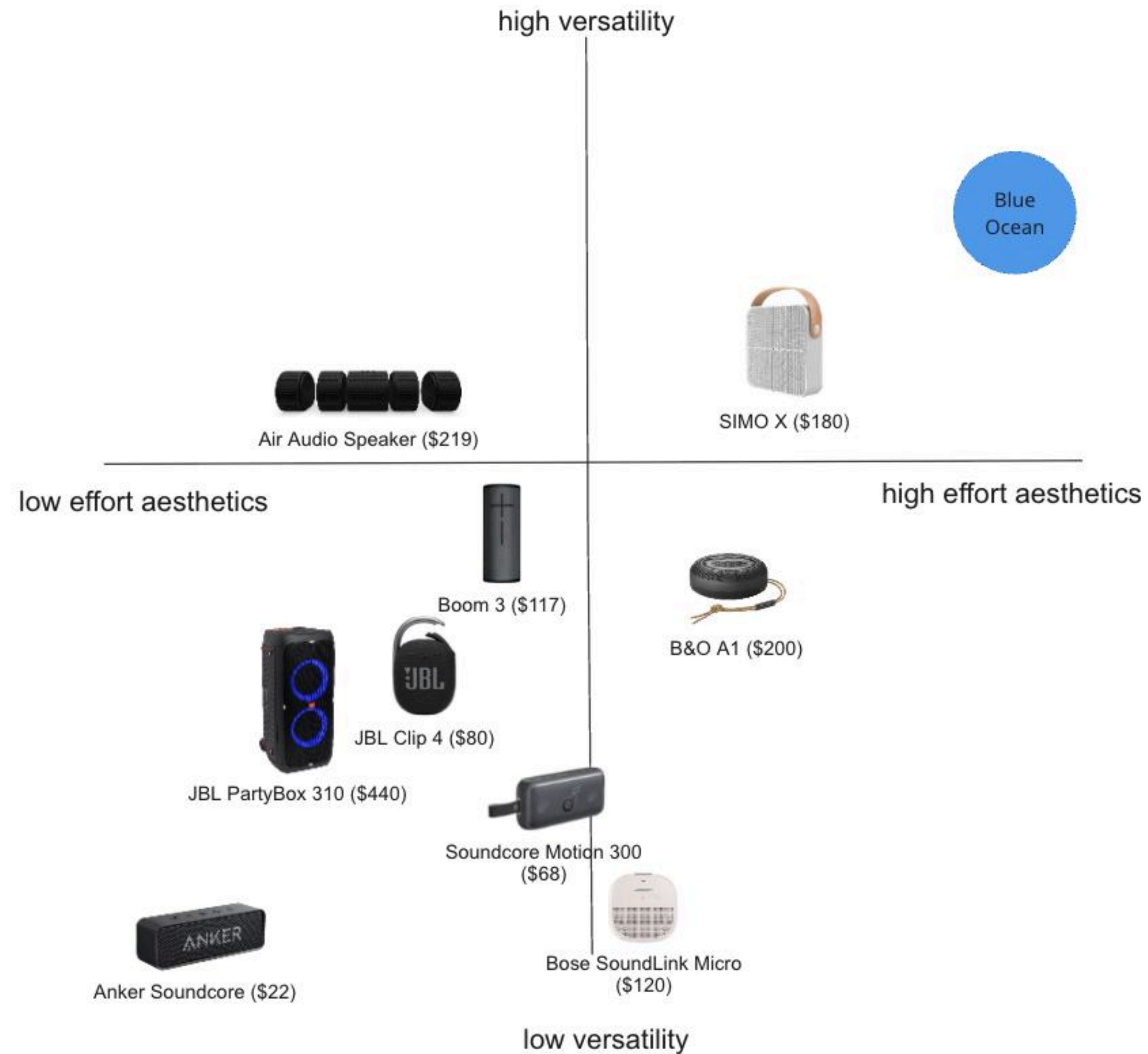
CONS: heavy, weak bass



# PRIMARY POSITIONING MATRIX

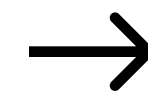


# SECONDARY POSITIONING MATRIX



## PROBLEM

our goal is to address limitations of current portable speakers, in terms of sound concentration and distribution. while compact speaker designs offer convenience, they tend to compromise on the immersive audio experience desired for group activities such as beach outings and hikes. the current market lacks a solution that seamlessly integrates both superior sound quality and the practicality of portability.



## OPPORTUNITY

as current offerings in the speaker market fall short in addressing the preferences of young adults, particularly those in college, we see an opportunity to create a speaker that prioritizes both portability and high sound quality. to meet this need, we aim to introduce a versatile speaker that seamlessly combines both elements, enhancing young adults' social and recreational experiences.



# DESIGN CRITERIA

**User-Friendly Setup** -  
intuitive  
pairing  
mechanism

**Portability** -  
Lightweight  
and  
convenient for  
storage

**Aesthetic** -  
Sleek and  
complements  
social settings.

**Personalization** -  
color choices &  
visual elements  
for added  
ambiance

**Durability**  
-Water,  
Sweat, and  
Dust Proof

**Modular** -  
interconnectivi  
ty between  
multiple  
speakers

**Affordable**  
- Cost-  
effective  
materials

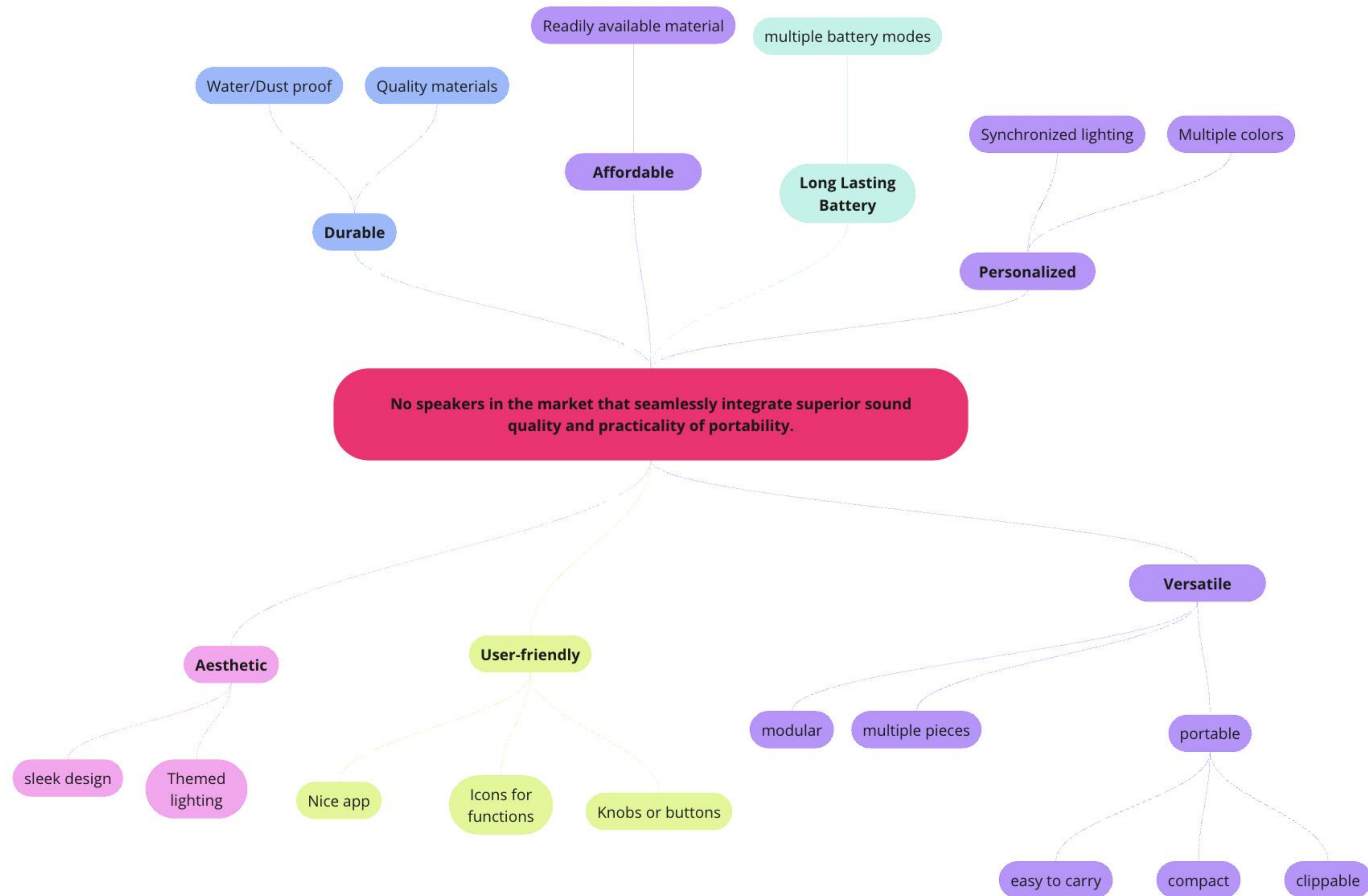
**Easy to Charge** - USB-  
C or wireless  
charging.

**Efficient power**  
- All Day Battery  
(>12 hours)

**Sound Quality** -  
speaker  
placement and  
acoustics



# MIND MAP



# FRAMING QUESTIONS

how might we  
make a smaller  
speaker with a  
better sound  
experience?

how might we  
create a portable  
speaker with good  
sound quality?

how might  
we make  
the speaker  
durable?

How might  
we make  
this more  
affordable?

how might we  
create a speaker  
thats both  
portable and  
aesthetic?

how might we  
incorporate  
versatility into  
an aesthetic  
design?

how might we  
redefine what  
it means for  
something to  
be portable?

how might  
we make  
the speaker  
aesthetic?

How might  
we make  
the speaker  
modular?

how might  
we maximize  
size and  
portability?



# WHAT-IF STATEMENTS





# Bluetooth Speaker redesigns



## IDEATION SKETCHES



# LOW-FIDELITY PROTOTYPE

## positives:

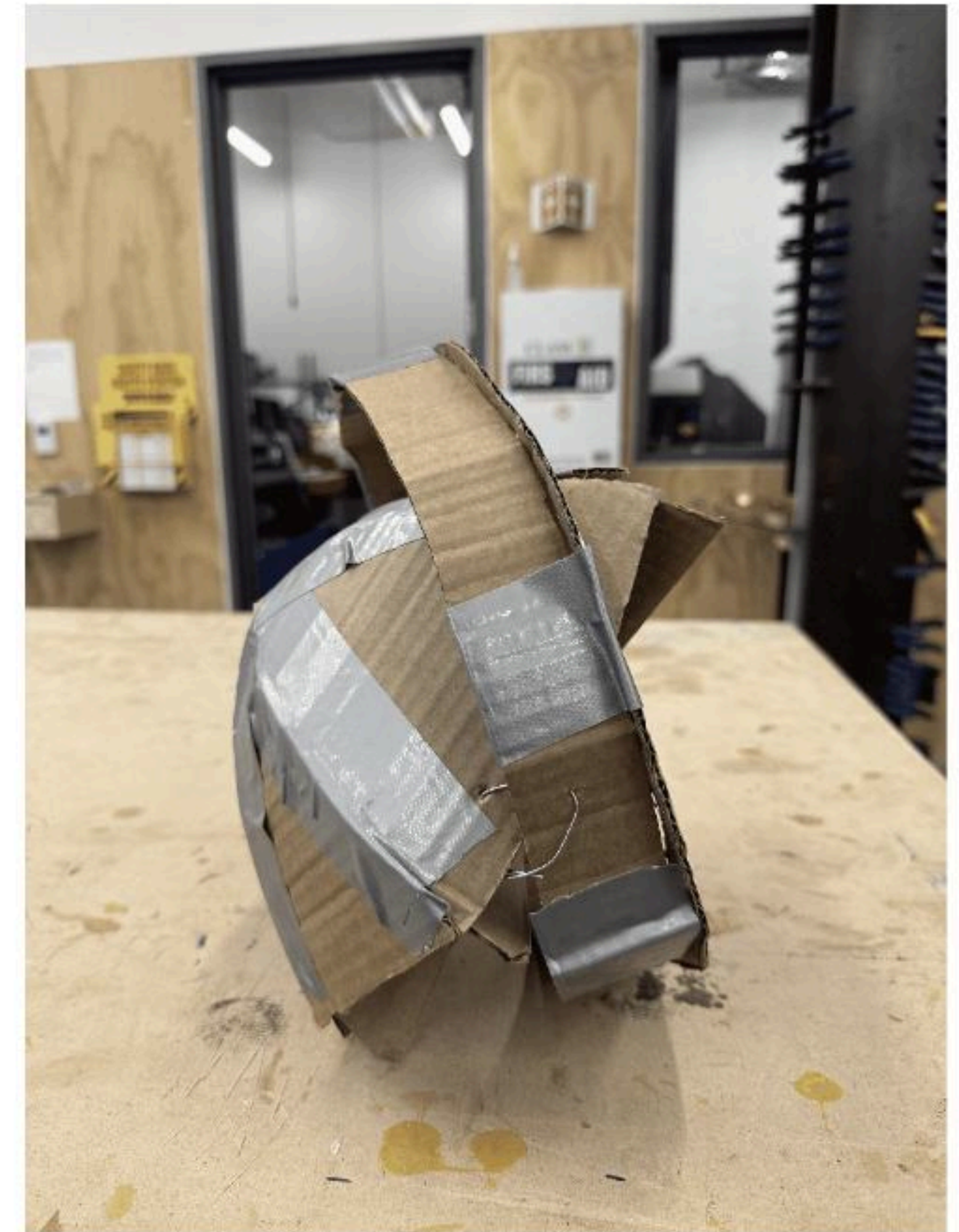
- this prototype was able to demonstrate the basic functionality of the speaker i was envisioning
- visualizing the functionality was my priority and i believe that was executed with this

## negatives:

- trying to make a very modular speaker with the cardboard was difficult, so aesthetically it was not aesthetically representative

## improvements:

- i want to find a way to improve the balance of the speaker
- a more modular model and smoother functionality would be ideal for my next prototype





# HIGH-FIDELITY PROTOTYPE

